Reasoning and Knowledge-based agents

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Lecture Objectives

- Types of Reasoning
- Agent modeling using Logic

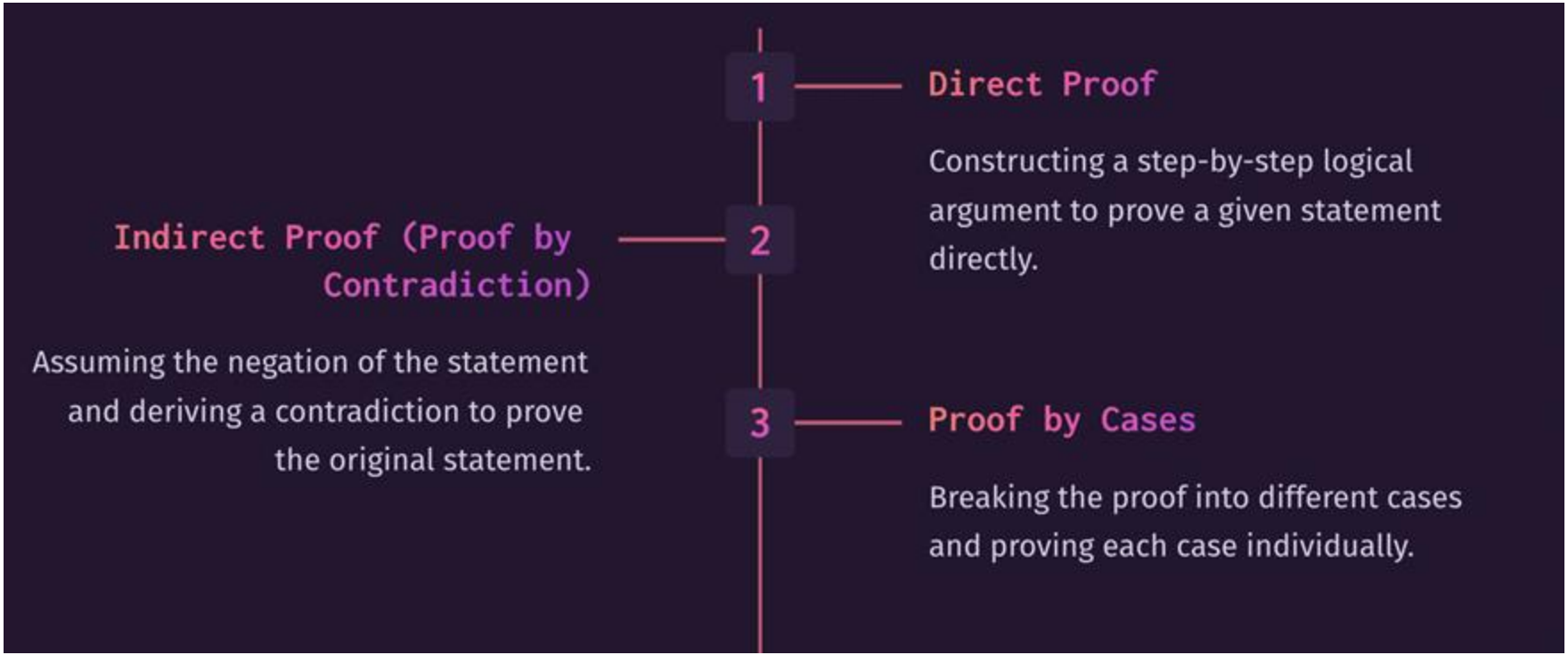
Types of reasoning

Deductive Reasoning: Using general statements to draw specific conclusions with certainty.

Inductive Reasoning: a method of drawing conclusions by going from the specific to the general. (bottom-up approach)

Counterexamples: Providing a specific example that disproves a universal statement.

Logic Proof Techniques



Induction: Example in Mathematics

Principle of Induction. Let P be any property of natural numbers.
Suppose P holds of zero, and whenever
P holds of a natural number n, then it holds of its successor, n+1.
Then P holds of every natural number.

Theorem. For every natural number n, $1+2+\ldots+2^n=2^{n+1}-1$.

Example in Mathematics (2)

Theorem. For every natural number n, $1+2+\ldots+2^n=2^{n+1}-1$.

Proof. We prove this by induction on n. In the base case, when $n = 0$, we have $1 = 2^{0+1} - 1$, as required.

For the induction step, fix n, and assume the *inductive hypothesis*

$$1 + 2 + \ldots + 2^n = 2^{n+1} - 1.$$

We need to show that this same claim holds with n replaced by $n + 1$. But this is just a calculation:

$$\begin{aligned} 1 + 2 + \ldots + 2^{n+1} &= (1 + 2 + \ldots + 2^n) + 2^{n+1} \\ &= 2^{n+1} - 1 + 2^{n+1} \\ &= 2 \cdot 2^{n+1} - 1 \\ &= 2^{n+2} - 1. \end{aligned}$$

Deduction Reasoning

In a simple deductive logic argument, you'll often begin with a premise, and add another premise. Then, you form a conclusion based on these two premises. This format is called "premise-premise-conclusion."

Here, A **proof** is an argument from **hypotheses** (assumptions) to a **conclusion**. Each step of the argument follows the laws of logic.

Axioms and theorems

- **Axioms** are facts and rules that attempt to capture all of the (important) facts and concepts about a domain; axioms can be used to prove **theorems**
  - Mathematicians don't want any unnecessary (dependent) axioms –ones that can be derived from other axioms
  - Dependent axioms can make reasoning faster, however
  - Choosing a **good set of axioms for a domain is a kind of design problem**

The Axioms of Euclidean Plane Geometry

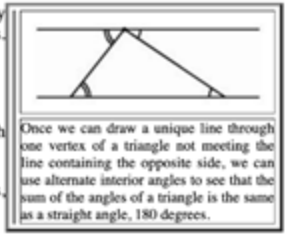
For well over two thousand years, people had believed that only one geometry was possible, and they had accepted the idea that this geometry described reality. One of the greatest Greek achievements was setting up rules for plane geometry. This system consisted of a collection of undefined terms like point and line, and five axioms from which all other properties could be deduced by a formal process of logic. Four of the axioms were so self-evident that it would be unthinkable to call any system a geometry unless it satisfied them:

1. A straight line may be drawn between any two points.
2. Any terminated straight line may be extended indefinitely.
3. A circle may be drawn with any given point as center and any given radius.
4. All right angles are equal.

But the fifth axiom was a different sort of statement:

5. If two straight lines in a plane are met by another line, and if the sum of the internal angles on one side is less than two right angles, then the straight lines will meet if extended sufficiently on the side on which the sum of the angles is less than two right angles.

Because this axiom was much more complicated than the previous axioms, it seemed more like a theorem than a self-evident proposition. Since all attempts to deduce it from the first four axioms had failed, Euclid simply included it as an axiom because he knew he needed it. For example, some axiom like this one was necessary for proving one of Euclid's most famous theorems, that the sum of the angles of a triangle is 180 degrees. Mathematicians found alternate forms of the axiom that were easier to state, for example:



Once we can draw a unique line through one vertex of a triangle not meeting the line containing the opposite side, we can use alternate interior angles to see that the sum of the angles of a triangle is the same as a straight angle, 180 degrees.

5'. For any given point not on a given line, there is exactly one line through the point that does not meet the given line.

This form of the fifth axiom became known as the parallel postulate. Although it was simpler to understand than Euclid's original formulation, it was no easier to deduce from the earlier axioms. The attempt to deduce the fifth axiom remained a great challenge right up to the nineteenth century, when it was proved that the fifth axiom did not follow from the first four.

The great advantage of expressing geometry as an axiomatic system was that it no longer was necessary to memorize long lists of independent facts about the nature of the universe--one only had to know a small set of axioms, and by applying to them the rules of inference, one could reconstruct the entire collection of geometric truths.

There was little doubt that the Greeks were attempting to describe a real world when they formulated their geometry, even though it might have been an ideal sort of world, realized only abstractly "in the mind of God." Many mathematicians, now as well as in the distant past, believe that the complete structure of mathematics is something that exists in itself and that it is only gradually discovered by human beings laboring to uncover its mysteries. Even though the framers of the early axiom systems would refer to point and line as undefined terms, they fairly clearly thought of them as real objects, and they thought that the system they were developing was a progressively more and more elaborate and accurate description of the real world. The progress of algebra, on the other hand, was not quite so settled, and people accepted changes in viewpoint there more readily than in the very traditional field of geometry.

Noncommutative Algebra

https://www.youtube.com/watch?v=LPET_HhN0VM

GENERAL CONCEPTS

- **Model:** an interpretation of a set of sentences where every sentence is *True. (a possible world).*
- E.g.having x men and y women sitting at a table playing bridge, and the sentence x + y = 4 is true when there are four people in total. Formally, the possible models are just all possible assignments of real numbers to the variables x and y. Each such assignment fixes the truth of any sentence of arithmetic whose variables are x and y. If a sentence α is true in model m, we say that m **satisfies** α or m **is a model of** α.

GENERAL CONCEPTS (2)

- **A sentence is**
  - **satisfiable** if it is true under some interpretation
  - **valid** if it is true under all possible interpretations
  - **inconsistent** if there does not exist any interpretation under which the sentence is true
- **Logical consequence:** S |= X if all models of S are also models of X
  - involves the relation of logical **entailment** between sentences—the idea that a sentence *follows logically* from another sentence.

Entailment and derivation

- **Entailment: KB |= Q**
  - Q is entailed by KB (a set of premises or assumptions) if and only if there is no logically possible world in which Q is false while all the premises in KB are true.

- **Derivation: KB |- Q**
  - We can derive Q from KB if there is a proof consisting of a sequence of valid inference steps starting from the premises in KB and resulting in Q

Two important properties for inference

Soundness: If KB |- Q then KB |= Q

- If Q is derived from a set of sentences KB using a given set of rules of inference, then Q is entailed by KB.
- **Soundness:** only true statements can be proven using the inference rules.

Completeness: If KB |= Q then KB |- Q

- If Q is entailed by a set of sentences KB, then Q can be derived from KB using the rules of inference.
- **Completeness** guarantees that we can prove everything that is true

Basic rules of inferences

Modus Ponens: If P implies Q, and P is true, then Q is true.

Modus Tollens: If P implies Q, and Q is false, then P is false.

Constructing a Conjunction: If P is true and Q is true, then P and Q is true.

Rule of Simplification: If P and Q is true, then P is true.

Rule of Addition: If P is true, then P or Q is true.

Rule of Disjunctive Syllogism: If P or Q is true, and P is false, then Q is true.

Rule of Hypothetical Syllogism: If P implies Q, and Q implies R, then P implies R.

Some of the rules of inference

- Each can be shown to be sound using a truth table

RULE	PREMISE	CONCLUSION
Modus Ponens	$A, A \rightarrow B$	B
And Introduction	A, B	$A \wedge B$
And Elimination	$A \wedge B$	A
Double Negation	$\neg\neg A$	A
Unit Resolution	$A \vee B, \neg B$	A
Resolution	$\mathbf{A \vee B, \neg B \vee C}$	$\mathbf{A \vee C}$

Example of Modus Ponens

Premise 1: $(A \vee \neg B) \rightarrow \neg C$

Premise 2: $\neg D$

Premise 3: $(A \vee \neg B)$

Example of Modus Ponens

Premise 1: $(A \vee \neg B) \rightarrow \neg C$

Premise 2: $\neg D$

Premise 3: $(A \vee \neg B)$

Conclusion: $\neg C$ (Modus Ponens (1,3))

Soundness of modus ponens

A	B	A → B	OK?
True	True	True	√
True	False	False	√
False	True	True	√
False	False	True	√

Soundness of the resolution inference rule

α	β	γ	$\alpha \vee \beta$	$\neg\beta \vee \gamma$	$\alpha \vee \gamma$
False	False	False	False	True	False
False	False	True	False	True	True
False	True	False	True	False	False
False	True	True	True	True	True
True	False	False	True	True	True
True	False	True	True	True	True
True	True	False	True	False	True
True	True	True	True	True	True

Exercise I

Premise 1: $\neg A \rightarrow (C \wedge D)$

Premise 2: $A \rightarrow B$.

Premise 3: $\neg B$

Prove: C.

Exercise II

Premise 1: ¬(A V B) → C

Premise 2: ¬A

Premise 3: ¬C

Prove: B.

Exercise III

Suppose we have the following scenario with a large knowledge base (KB) consisting of the following statements:

1. If it is raining, then the streets are wet.
2. If the streets are wet, then people use umbrellas.
3. It is raining.
4. All dogs are mammals.
5. All mammals have hair.
6. Some animals have fur.
7. All birds have feathers.
8. Penguins are birds.
9. Penguins cannot fly.
10. All fruits have seeds.
11. Apples are fruits.
12. Oranges are fruits.
13. Bananas are fruits.
14. All vegetables are healthy.
15. Spinach is a vegetable.
16. Carrots are vegetables.
17. Broccoli is a vegetable.
18. All cars have wheels.
19. A Tesla is a car.
20. Teslas are electric vehicles.

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Prove the statements:

- People are using umbrellas because it is raining.
- Penguins have feathers.
- Spinach is healthy.
- All dogs have hair.
- Penguins cannot fly.

Exercise IV

If the unicorn is mythical, then it is immortal, but if it is not mythical, then it is a mortal mammal. If the unicorn is either immortal or a mammal, then it is horned. The unicorn is magical if it is horned.

Given the above, can you prove that the unicorn is mythical? How about magical? Horned?

Solution.

Propositional logic statements of the given facts:

(1) Mythical → ¬Mortal

(2) ¬Mythical → Mortal ∧Mammal

(3) ¬Mortal ∨ Mammal → Horned

(4) Horned → Magical

We show how inference is done by direct application.

(5) ¬Mythical ∨ ¬Mortal From (1)

(6a) (Mythical ∨ Mortal) From (2)

(6b) (Mythical ∨ Mammal) From (2)

(7) ¬Mortal ∨ Mammal Resolution on (5) and (6b)

(8) Horned Modus Ponens on (3) and (7)

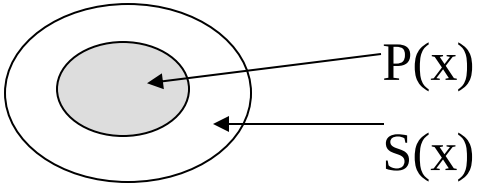
(9) Magical Modus Ponens on (4) and (8) Thus the unicorn is both magical and horned.

Necessary and sufficient conditions

- A **necessary** condition must be satisfied for a statement to be true.
- A **sufficient** condition, if satisfied, assures the statement's truth.
- Duality: "P is sufficient for Q" is the same as "Q is necessary for P."

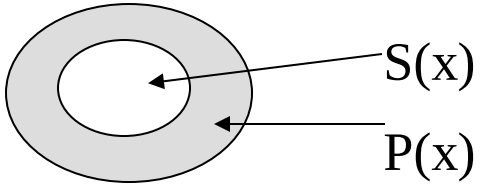
Necessary and sufficient conditions

S(x) is a necessary condition of P(x)



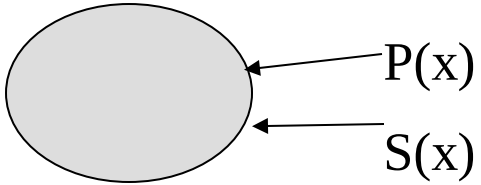
$(\forall x)\ P(x) \Rightarrow S(x)$

S(x) is a sufficient condition of P(x)



$(\forall x)\ P(x) \Leftarrow S(x)$

S(x) is a necessary and sufficient condition of P(x)



$(\forall x)\ P(x) \Leftrightarrow S(x)$

Common mistakes to avoid in Rules of Inference and Logic Proofs

1. **Starting with a false premise:** If you start with a false premise, you can prove anything. It is important to ensure that your premises are true before proceeding with the proof.
2. **The fallacy of equivocation:** Errors can result from using undefined symbols, misplacing parentheses, or using symbols incorrectly. Using a term with multiple meanings inconsistently throughout an argument.
3. **The fallacy of affirming the consequent:** Using the wrong rule of inference can lead to an incorrect conclusion. It is important to understand the different rules of inference and when to use them.
4. **Failing to justify each step:** Each step of the argument must follow the laws of logic. It is important to justify each step of the proof to ensure that it is valid.
5. **Making errors in form:** When dealing with arguments in natural language, making errors in form is quite common. It is important to be familiar with the different forms of arguments and to ensure that the form of your argument is valid.

Summary

- The process of deriving new sentences from old one is called **inference**.
  - **Sound** inference processes derives true conclusions given true premises
  - **Complete** inference processes derive all true conclusions from a set of premises
- A **valid sentence** is true in all worlds under all interpretations
- If an implication sentence can be shown to be valid, then—given its premise—its consequent can be derived

Summary

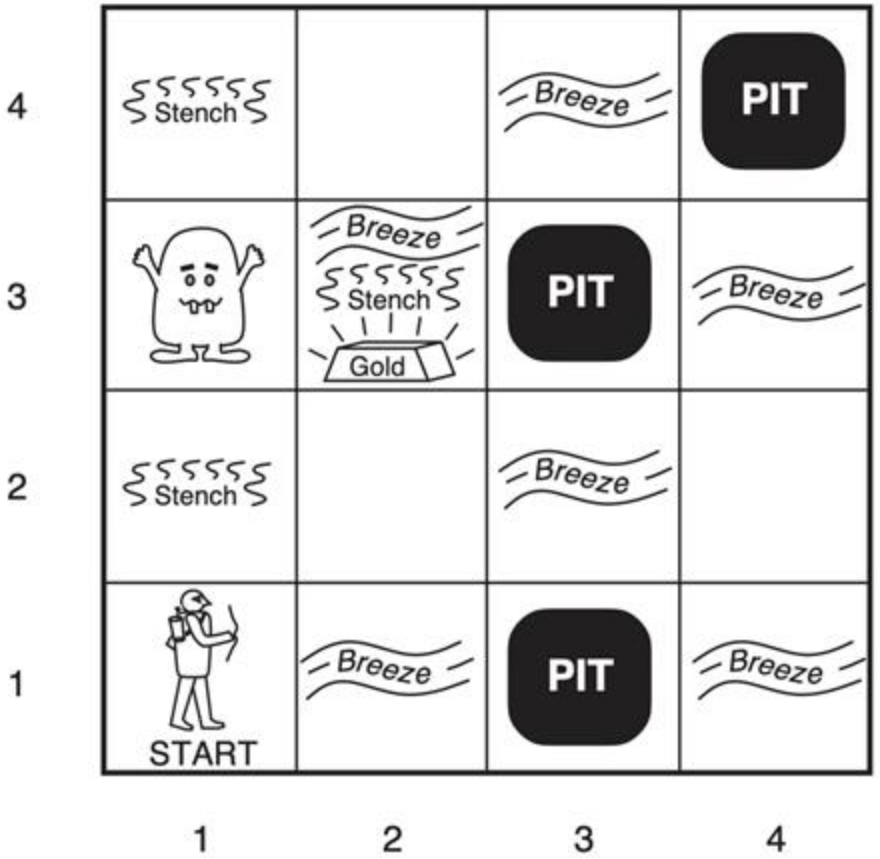
- Different logics make different **commitments** about what the world is made of and what kind of beliefs we can have regarding the facts
- **Propositional logic** commits only to the existence of facts that may or may not be the case in the world being represented
  - It has a simple syntax and simple semantics. It suffices to illustrate the process of inference
  - Propositional logic quickly becomes impractical, even for very small worlds

Logical agents

Three (non-exclusive) agent architectures:

- Reflex agents
  - Have rules that classify situations, specifying how to react to each possible situation
- Model-based agents
  - Construct an internal model of their world
- Goal-based agents
  - Form goals and try to achieve them

Wumpus World



A simple reflex agent

- Rules to map percepts into observations:

  $\forall b,g,u,c,t\ \text{Percept}([\text{Stench}, b, g, u, c], t) \rightarrow \text{Stench}(t)$

  $\forall s,g,u,c,t\ \text{Percept}([s, \text{Breeze}, g, u, c], t) \rightarrow \text{Breeze}(t)$

  $\forall s,b,u,c,t\ \text{Percept}([s, b, \text{Glitter}, u, c], t) \rightarrow \text{AtGold}(t)$

- Rules to select an action given observations:

  $\forall t\ \text{AtGold}(t) \rightarrow \text{Action}(\text{Grab}, t);$

- Some difficulties:
  - **Consider Climb.** There is no percept that indicates the agent should climb out – position and holding gold are not part of the percept sequence
  - **Loops** – the percept will be repeated when you return to a square, which should cause the same response (unless we maintain some internal model of the world)

The “Hunt the Wumpus” agent

- Some atomic propositions:

  S12 = There is a stench in cell (1,2)

  B34 = There is a breeze in cell (3,4)

  W22 = The Wumpus is in cell (2,2)

  V11 = We have visited cell (1,1)

  OK11 = Cell (1,1) is safe.

  etc

- Some rules:

  (R1) $\neg S11 \rightarrow \neg W11 \wedge \neg W12 \wedge \neg W21$

  (R2) $\neg S21 \rightarrow \neg W11 \wedge \neg W21 \wedge \neg W22 \wedge \neg W31$

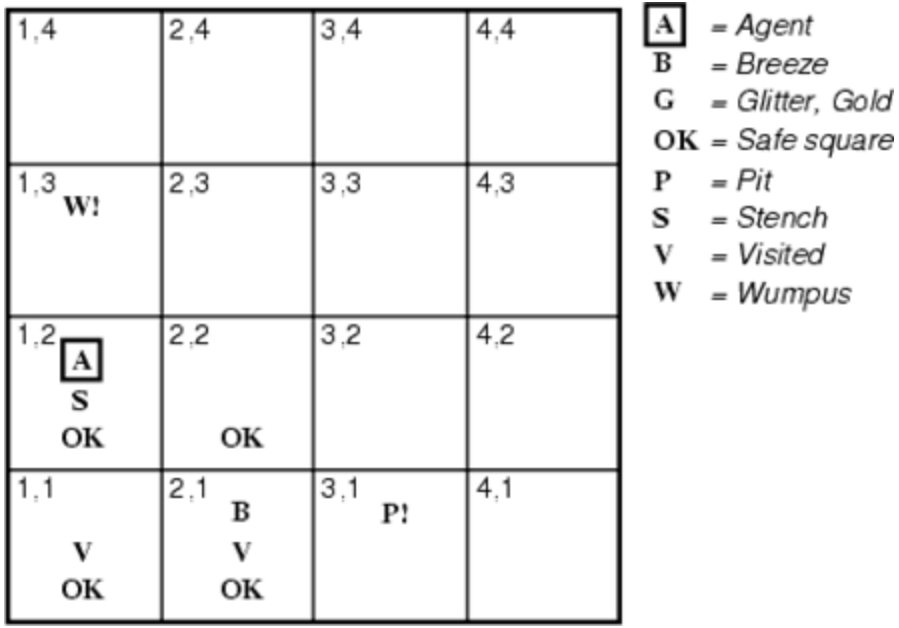
  (R3) $\neg S12 \rightarrow \neg W11 \wedge \neg W12 \wedge \neg W22 \wedge \neg W13$

  (R4) $S12 \rightarrow W13 \vee W12 \vee W22 \vee W11$

  etc

After the third move

- We can prove that the Wumpus is in (1,3) using the four rules given.



Proving W13

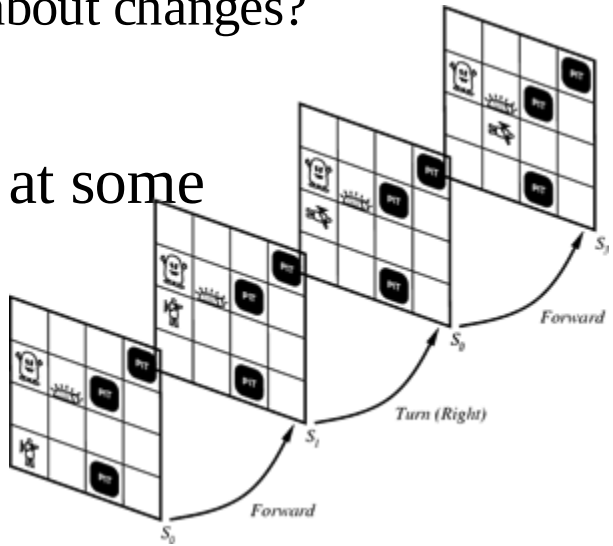
- Apply MP with ¬S11 and R1:
  ¬ W11 ∧ ¬ W12 ∧ ¬ W21
- Apply And-Elimination to this, yielding 3 sentences:
  ¬ W11, ¬ W12, ¬ W21
- Apply MP to ~S21 and R2, then apply And-elimination:
  ¬ W22, ¬ W21, ¬ W31
- Apply MP to S12 and R4 to obtain:
  W13 ∨ W12 ∨ W22 ∨ W11
- Apply Unit resolution on (W13 ∨ W12 ∨ W22 ∨ W11) and ¬W11:
  W13 ∨ W12 ∨ W22
- Apply Unit Resolution with (W13 ∨ W12 ∨ W22) and ¬W22:
  W13 ∨ W12
- Apply UR with (W13 ∨ W12) and ¬W12:
  W13

Problems with the propositional Wumpus hunter

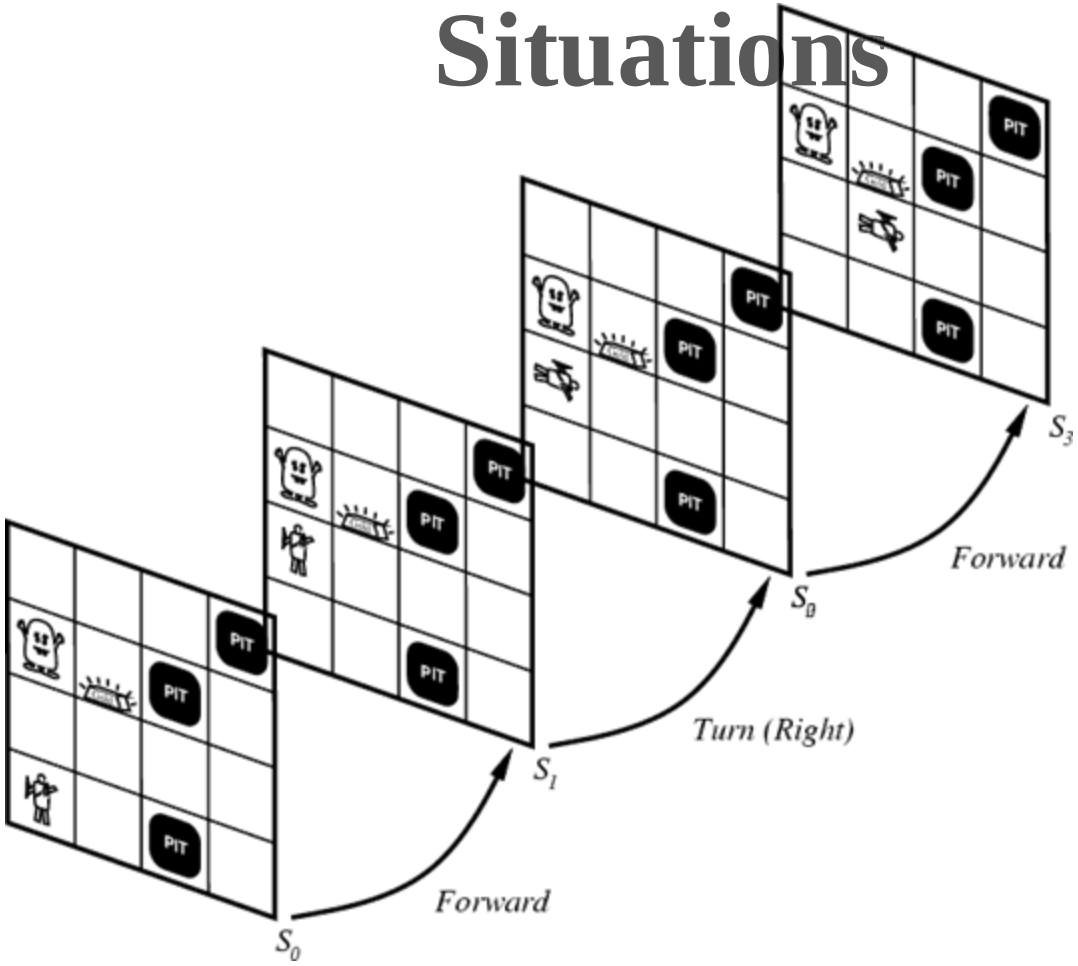
- Lack of variables prevents stating more general rules
  - We need a set of similar rules for each cell
- Change of the KB over time is difficult to represent
  - Standard technique is to index facts with the time when they’re true
  - This means we have a separate KB for every time point

Representing change

- Representing change in the world in logic can be tricky.
- One way is just to change the KB
  - Add and delete sentences from the KB to reflect changes
  - How do we remember the past, or reason about changes?
- **Situation calculus** is another way
- A **situation** is a snapshot of the world at some instant in time
- When the agent performs an action A in situation S1, the result is a new situation S2.



Situations


PIT
PIT
PIT
S_0
Forward
PIT
PIT
PIT
S_1
Turn (Right)
PIT
PIT
PIT
S_0
Forward
S_3

Situation calculus

- A **situation** is a snapshot of the world at an interval of time during which nothing changes
- Every true or false statement is made with respect to a particular situation.
  - Add **situation variables** to every predicate.
  - at(Agent,1,1) becomes at(Agent,1,1,s0): at(Agent,1,1) is true in situation (i.e., state) s0.
  - Alternatively, add a special 2nd-order predicate, **holds(f,s),** that means "f is true in situation s." E.g., holds(at(Agent,1,1),s0)
- Add a new function, **result(a,s),** that maps a situation s into a new situation as a result of performing action a. For example, result(forward, s) is a function that returns the successor state (situation) to s
- Example: The action agent-walks-to-location-y

Qualification problem

- Qualification problem:
  - How can you possibly characterize every single effect of an action, or every single exception that might occur?
  - When I put my bread into the toaster, and push the button, it will become toasted after two minutes, unless…
    - The toaster is broken, or…
    - The power is out, or…
    - I blow a fuse, or…
    - A neutron bomb explodes nearby and fries all electrical components, or…
    - A meteor strikes the earth, and the world we know it ceases to exist, or…

Ramification problem

- Similarly, it's just about impossible to characterize every side effect of every action, at every possible level of detail:
  - When I put my bread into the toaster, and push the button, the bread will become toasted after two minutes, and…
    - The crumbs that fall off the bread onto the bottom of the toaster over tray will also become toasted, and…
    - Some of the aforementioned crumbs will become burnt, and…
    - The outside molecules of the bread will become "toasted," and…
    - The inside molecules of the bread will remain more "breadlike," and…
    - The toasting process will release a small amount of humidity into the air because of evaporation, and…
    - The heating elements will become a tiny fraction more likely to burn out the next time I use the toaster, and…
    - The electricity meter in the house will move up slightly, and…

Knowledge engineering!

- Modeling the “right” conditions and the “right” effects at the “right” level of abstraction is very difficult
- Knowledge engineering (creating and maintaining knowledge bases for intelligent reasoning) is an entire field of investigation
- Many researchers hope that automated knowledge acquisition and machine learning tools can fill the gap:
  - Our intelligent systems should be able to **learn** about the conditions and effects, just like we do!
  - Our intelligent systems should be able to learn when to pay attention to, or reason about, certain aspects of processes, depending on the context!

Preferences among actions

- A problem with Wumpus world knowledge base that we have built so far is that it is difficult to decide which action is best among a number of possibilities.
- For example, to decide between a forward and a grab, axioms describing when it is OK to move to a square would have to mention glitter.
- This is not modular!
- We can solve this problem by **separating facts about actions from facts about goals**. This way our **agent can be reprogrammed just by asking it to achieve different goals**.

Preferences among actions

- The first step is to describe the desirability of actions independent of each other.
- In doing this we will use a simple scale: actions can be Great, Good, Medium, Risky, or Deadly.
- Obviously, the agent should always do the best action it can find:

  $(\forall a,s)\ \text{Great}(a,s) \rightarrow \text{Action}(a,s)$

  $(\forall a,s)\ \text{Good}(a,s) \wedge \neg(\exists b)\ \text{Great}(b,s) \rightarrow \text{Action}(a,s)$

  $(\forall a,s)\ \text{Medium}(a,s) \wedge (\neg(\exists b)\ \text{Great}(b,s) \vee \text{Good}(b,s)) \rightarrow \text{Action}(a,s)$

  ...

Preferences among actions

- We use this action quality scale in the following way.
- Until it finds the gold, the basic strategy for our agent is:
  - Great actions include picking up the gold when found and climbing out of the cave with the gold.
  - Good actions include moving to a square that’s OK and hasn't been visited yet.
  - Medium actions include moving to a square that is OK and has already been visited.
  - Risky actions include moving to a square that is not known to be deadly or OK.
  - Deadly actions are moving into a square that is known to have a pit or a Wumpus.


Q & A